Simon Chen

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Education

Boston University — Boston, MA
Bachelor of Arts in Computer Science — *Graduated: Spring 2026*

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Databases, Machine Learning, Data Science.

Experience

Product Lead — December 2025 – Present
Ezesports — *New York, NY*

- Migrated an esports league platform from Google Sites to Next.js with a Supabase backend, replacing manual content updates with live scoring and automated season management.
- Designed a PostgreSQL schema on Supabase using Drizzle ORM, supporting 500+ matches across 20+ teams, and built RESTful endpoints for player statistics, standings, and historical data.
- Cut page load time by 75% (from 3.2s to 0.8s) by implementing edge caching on Cloudflare, route-based code splitting, and asset optimization for the Next.js frontend.

Software Engineering Intern — February 2026 – June 2026
RESET Standard — *Shanghai, China*

- Doubled the platform's supported environmental data categories (from 2 to 4) by independently implementing end-to-end water and waste data integration, developing Rails backend ingestion pipelines, PostgreSQL schemas, and client-facing React/Vite/Tailwind dashboards.
- Engineered asynchronous data pipelines with RabbitMQ, Redis, and Sidekiq to process long-running ingestion jobs in the background, decoupling external API calls from user-facing requests.
- Redesigned project-device mapping workflows in response to client requests, refactoring legacy forms to streamline onboarding of environmental monitoring devices.

IT Consultant — January 2024 – July 2025
Boston University Engineering IT — *Boston, MA*

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions on shared NAS storage, conducting recurring access audits
- Enhanced endpoint security with CrowdStrike across 50+ devices, resolving 20+ support tickets per week

Projects

Hermes Letters | *Next.js, TypeScript, Supabase, Drizzle ORM, PostgreSQL* — June 2026

- Built a private long-form messaging platform for a group of 20+ friends during a study abroad semester, delivering 30+ letters.
- Ensured consistent letter claims via atomic SQL updates and secured sessions in XSS-resistant httpOnly cookies.
- Derived user connections at query time from kept-letter records using indexed lookups, avoiding a separate relationship table and associated database sync issues.
- Hardened media storage by validating uploads via server-side magic-byte signatures, storing files in a private Supabase bucket, and restricting delivery to 1-hour signed URLs.

Persephone | *Python, Cython/C++17, Rust, Modal, FAISS* — May 2026

- Built and operated a live Kalshi BTC prediction-market trading system end to end (market-data ingestion, theo serving, strategy evaluation, execution, and risk), generating $8K net PnL on $200K traded volume in its first 4 days.
- Replaced KDE inference with a GBM theo layer, cutting model query latency from $\sim$1.5ms to $\sim$20μs per query ($\sim$75x) behind an unchanged live serving interface.
- Cut backtest and parameter-sweep iteration from $\sim$14 hours on laptop CPUs to $\sim$15 minutes by moving tick backtests and sweeps to Modal NVIDIA GPU workers with FAISS GPU search and CuPy-vectorized math.
- Sped up 155K-row neighbor search from 758μs to 154μs p50 by caching FAISS IVF indexes and pinning FAISS threads to avoid OpenMP oversubscription.

- Rewrote 9 live hot-path kernels (order-book maintenance, order-book imbalance, trade intensity, event aggregation) as native Cython extensions, cutting Kalshi book maintenance from 21.8ms to 195μs p50 (112x).
- Designed a lock-free SPSC event ring in Cython/C++17 with cache-line-aligned atomic cursors and packed uint64 records, benchmarked at 584ns per enqueue and 703ns per event drained.
- Prevented bad-price quoting with fail-closed gates that block stale, crossed, or untrusted market data before quote generation, computing fused market metrics in 357μs p50 across 8 venues.
- Eliminated duplicate-send and crash-replay risk with a durable-before-wire order layer: intent journaled before dispatch, deterministic client IDs for idempotent retries, and persisted ack state for reconciliation.

Self-Hosted Server Infrastructure | *Docker, Linux, Cloudflare, Nginx, Automation* November 2025

- Architected containerized home-server infrastructure using Docker Compose, orchestrating 10+ self-hosted services for media streaming, automated library management, and monitoring
- Implemented secure external access using Cloudflare Tunnels with zero-trust network architecture, eliminating port forwarding vulnerabilities while hosting personal website and services accessible via custom domain
- Configured automated workflow pipelines with inter-container networking and Nginx reverse proxy routing, establishing centralized monitoring for resource optimization

Audio Transcription Platform / Whisperrr | *Java Spring Boot, Python FastAPI, React, PostgreSQL / Tailwind CSS*Septer

- Built full-stack transcription platform processing audio at 32x realtime speed, supporting 99+ languages across multiple audio formats with real-time results
- Designed three-tier stateless architecture with HTTP connection pooling to reduce request latency by 70%, improving reliability for files over 25MB from 60% to 100% success rate
- Containerized microservices with Docker Compose health checks and implemented segment-level timestamp generation for subtitle creation

Fitness Tracking Mobile App | *React Native, TypeScript, Expo* August 2025

- Built cross-platform app with drag-and-drop workout creation using React Native Reanimated v3 and PanGestureHandler for exercise reordering
- Implemented persistent data storage with AsyncStorage and custom React hooks for state management across workout sessions
- Developed modular component architecture with search/filter functionality and modal-based exercise selection interface

Food Waste Platform (BU Spark!) | *Django, JS, Auth0* September 2024 – December 2024

- Developed food waste reduction platform connecting students with leftover event food, implementing dual authentication (Django + Auth0 OAuth), QR code email delivery system, and many-to-many reservation management with real-time capacity tracking
- Integrated Google Maps JavaScript API for interactive event discovery with user geolocation and distance calculations, implementing multi-criteria filtering across 18 food categories and 8 allergen types using Django class-based views with form validation
- Deployed production Django application with WhiteNoise static file serving and automated Gmail SMTP notifications featuring inline QR code attachments, managing SQLite database with 100+ campus events while implementing comprehensive security features including CSRF protection and input validation

Crime Analytics & Forecasting Platform / Crime-Mapper Boston | *Python, Streamlit, Prophet, Scikit-learn, GeoPandas*

- Developed crime forecasting system analyzing 204,356 incidents across 14 Boston districts using Facebook Prophet time series model, achieving up to 59.9% improvement over naive baseline forecasts
- Implemented automated model evaluation comparing Prophet against 3 baseline methods (seasonal naive, moving average, linear regression) with reliability indicators when simpler models perform better
- Deployed interactive dashboard with DBSCAN spatial clustering for hotspot identification, automatic API data refresh, and 2-month crime forecasts with confidence intervals

Decentralized ML Model Marketplace | *Solidity, Circom, IPFS, Express.js, TensorFlow* May 2024

- Built decentralized marketplace using Solidity smart contracts with ERC-20 tokens, implementing 6-state transaction management, automated candidate selection, and economic incentive mechanisms with stake slashing
- Integrated IPFS distributed storage with Ethereum blockchain using Thirdweb SDK, enabling secure model uploads with SHA256 hash verification and automated ML training workflows
- Implemented zero-knowledge proof system using Circom and Groth16 protocol for privacy-preserving accuracy verification, with MetaMask integration and real-time blockchain event monitoring

Technical Skills

Languages: Python, C++17, Cython, Rust, TypeScript, JavaScript, Java, Ruby, SQL
Frameworks & Libraries: React, Next.js, Node.js, React Native, Ruby on Rails, Django, Spring Boot, FastAPI, Tailwind CSS, FAISS, CuPy
Databases & Infrastructure: PostgreSQL, SQLite, Redis, RabbitMQ, Sidekiq, Supabase, Docker, Nginx, Cloudflare, Modal, OpenMP